

Module -2

General Second Order Circuits

Dr. Sathya P
Associate Professor
SENSE

Steps to solve general second order circuits

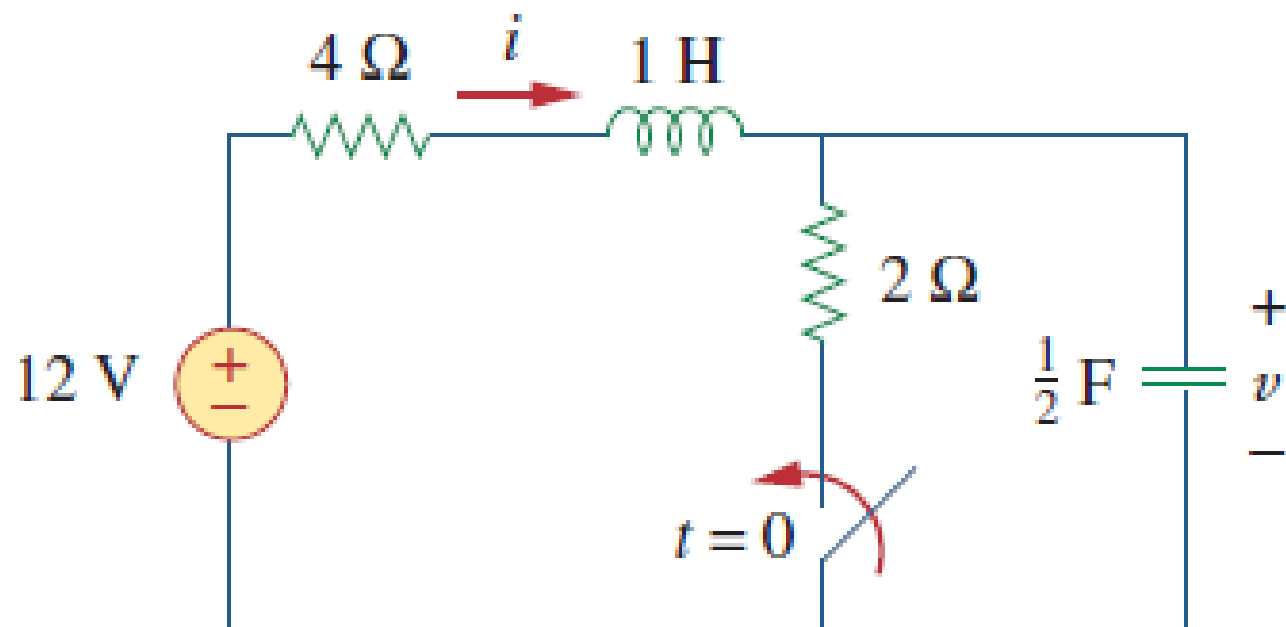
- Determine the initial conditions $x(0)$, $dx(0)/dt$ and final value $x(\infty)$.
- Turn off the independent sources and find the form of transient response $x_t(t)$ by applying KCL and KVL.
- Obtain the second order differential equation using KVL and KCL from step 2.
- Find the characteristic roots. Depending on the damping condition, write the transient equation and obtain the constants.
- The total response is sum of transient response and steady state response.

$$x(t) = x_t(t) + x_{ss}(t)$$

$$x_{ss}(t) = x(\infty)$$

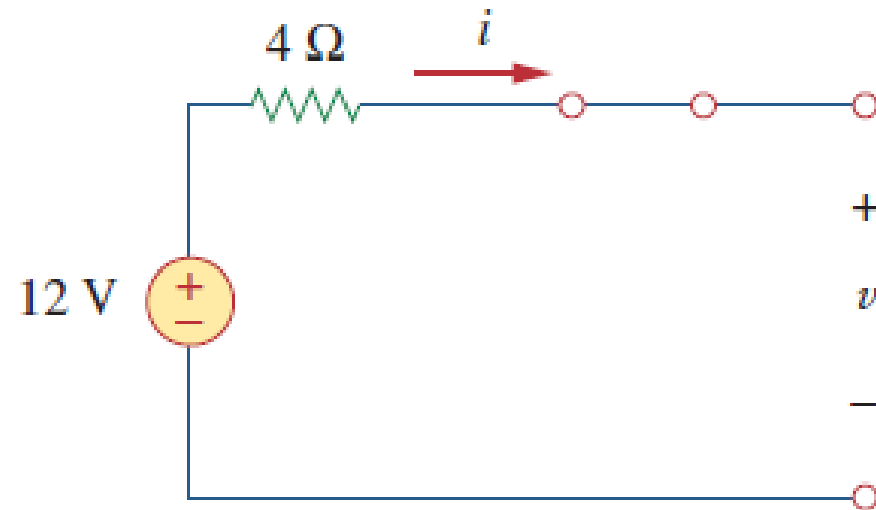
Numerical - 1

Find the complete response v and then i for $t > 0$ in the circuit



Solution:

We first find the initial and final values. At $t = 0^-$, the circuit is at steady state. The switch is open;



$$v(0^-) = 12 \text{ V}, \quad i(0^-) = 0$$

At $t = 0^+$, the switch is closed;

$$v(0^+) = v(0^-) = 12 \text{ V}, \quad i(0^+) = i(0^-) = 0$$

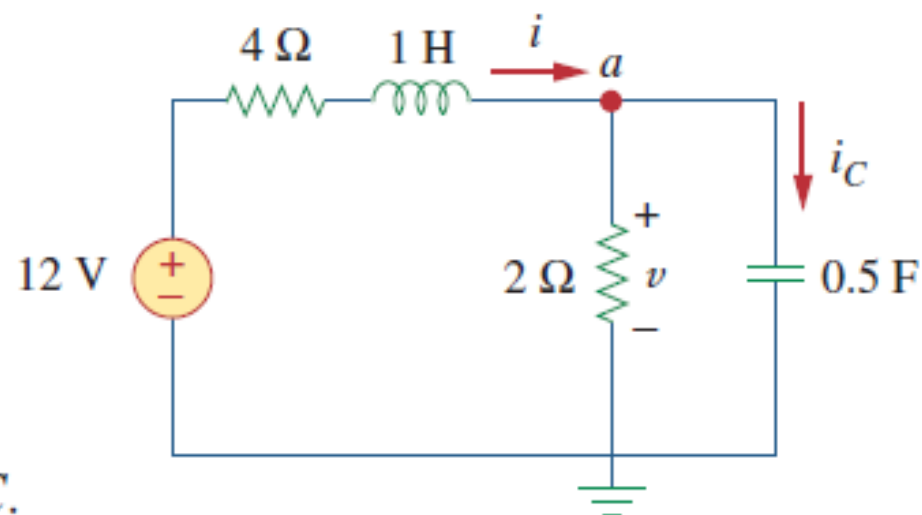
To get $dv(0^+)/dt$, we use $C dv/dt = i_C$ or $dv/dt = i_C/C$.

KCL at node a

$$i(0^+) = i_C(0^+) + \frac{v(0^+)}{2}$$

$$0 = i_C(0^+) + \frac{12}{2} \quad \Rightarrow \quad i_C(0^+) = -6 \text{ A}$$

$$\frac{dv(0^+)}{dt} = \frac{-6}{0.5} = -12 \text{ V/s}$$



The final values are obtained when the inductor is replaced by a short circuit and the capacitor by an open circuit

$$i(\infty) = \frac{12}{4 + 2} = 2 \text{ A}, \quad v(\infty) = 2i(\infty) = 4 \text{ V}$$

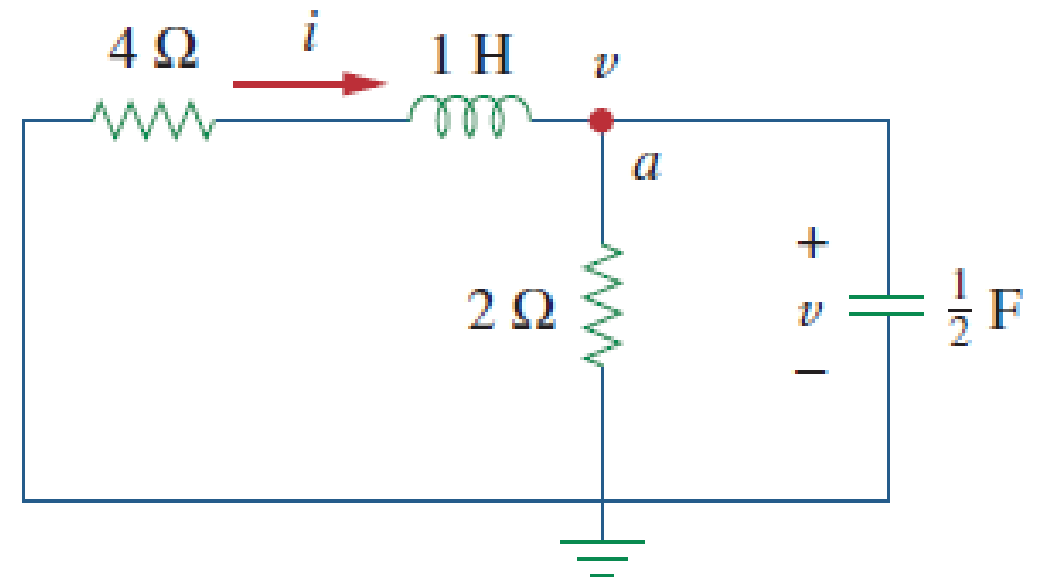
Next, we obtain the form of the transient response for $t > 0$. By turning off the 12-V voltage source,

Applying KCL at node a

$$i = \frac{v}{2} + \frac{1}{2} \frac{dv}{dt}$$

Applying KVL to the left mesh results in

$$4i + 1 \frac{di}{dt} + v = 0$$



$$2v + 2\frac{dv}{dt} + \frac{1}{2}\frac{dv}{dt} + \frac{1}{2}\frac{d^2v}{dt^2} + v = 0$$

$$\frac{d^2v}{dt^2} + 5\frac{dv}{dt} + 6v = 0$$

From this, we obtain the characteristic equation as

$$s^2 + 5s + 6 = 0$$

with roots $s = -2$ and $s = -3$. Thus, the natural response is

$$v_n(t) = Ae^{-2t} + Be^{-3t}$$

$$v_{ss}(t) = v(\infty) = 4$$

The complete response is

$$v(t) = v_t + v_{ss} = 4 + Ae^{-2t} + Be^{-3t}$$

We now determine A and B using the initial values.

$$12 = 4 + A + B \quad \Rightarrow \quad A + B = 8$$

$$\frac{dv}{dt} = -2Ae^{-2t} - 3Be^{-3t} \quad A = 12, \quad B = -4$$

$$v(t) = 4 + 12e^{-2t} - 4e^{-3t} \text{ V}, \quad t > 0$$

$$\begin{aligned} i &= \frac{v}{2} + \frac{1}{2} \frac{dv}{dt} = 2 + 6e^{-2t} - 2e^{-3t} - 12e^{-2t} + 6e^{-3t} \\ &= 2 - 6e^{-2t} + 4e^{-3t} \text{ A}, \quad t > 0 \end{aligned}$$